A Conditional Switching Roadmap for the Mean-Zero Version of Erdős Problem #521

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Abstract

This note records a conditional switching-disproof architecture for the mean-zero Rademacher interpretation of Erdős Problem #521. It does not address the literal non-centered $\{0,1\}$ formulation. The claim is conditional: if the local template, finite-window linear algebra, conditional anti-concentration, and off-core stability lemmas listed below can be proved with the stated exponents, then the almost-sure statement $R_n/\log n \to 2/\pi$ fails in the mean-zero $\{-1,1\}$ model.

1 Scope

We work with iid Rademacher coefficients

$$f_n(x) = \sum_{k=0}^{n} \epsilon_k x^k, \qquad \epsilon_k \in \{-1,1\}.$$

Let $R_n(f)$ denote the number of real roots. The target under discussion is the almost-sure assertion

$$\frac{R_n(f)}{\log n} \to \frac{2}{\pi}.$$

The inside interval is not the live obstruction: Do's theorem gives the almost-sure law

$$N_n([-1,1])/\log n \to 1/\pi$$

for iid zero-mean, unit-variance coefficients with bounded $(2+\eta)$-moment, hence for Rademacher coefficients. The delicate part is the exterior contribution, because the reciprocal symmetry that works at fixed n reverses coefficient time and does not automatically give an almost-sure statement along a single infinite sequence.

The switching direction below follows the switching and moment-matching strategy suggested by Terry Tao in earlier comments on the problem.

2 Coupling Contradiction Criterion

Assume, for contradiction, that $R_n(f)/\log n \to 2/\pi$ almost surely. Then for any sequence of dyadic couplings π_N of two iid Rademacher marginal sequences f, f',

$$\mathbb{P}_{\pi_N}\left(\sup_{N\le m\le 2N} |R_m(f) - R_m(f')| > \varepsilon \log N\right) \to 0$$

for every fixed $\varepsilon > 0$. This is just the triangle inequality on the common tail event where both marginals satisfy the assumed almost-sure convergence. The coupling may depend on N, and only the two marginal laws are used.

Thus it is enough to construct dyadic couplings preserving the iid marginals such that, for some fixed $c > 0$,

$$\mathbb{P}_{\pi_N}\left(\sup_{N\le m\le 2N} |R_m(f) - R_m(f')| \ge c\log N\right) \to 1.$$

3 Stationary Fiber Switching

Choose parameters

$$0 < a < b < \chi < 1, \qquad R_0 = N^a, \qquad R_1 = N^b, \qquad \rho_N = R_0/R_1 = N^{a-b}.$$

There are $N^{1-\chi+o(1)}$ checkpoints $n_j \in [N, 2N]$, separated by about N^χ. Around n_j we switch a late-coefficient block

$$n_j - r, \qquad R_0 \le r \le R_1.$$

On each block I, let

$$T_I : \{\pm 1\}^I \to \mathbb{Z}^d$$

be the fixed finite ordinary/alternating guard-moment map. Given an input block x, sample the output block uniformly from the fiber

$$T_I^{-1}(T_I(x)).$$

This kernel is doubly stochastic. Indeed, for every fixed output block y,

$$\sum_{x:T_I(x)=T_I(y)} 2^{-|I|}\frac{1}{|T_I^{-1}(T_I(x))|} = 2^{-|I|}.$$

Applying these kernels independently on disjoint blocks preserves the full iid Rademacher product law, including independence across output blocks and untouched coordinates.

The fixed map T_I only preserves finite guard moments. The $M \asymp \log N$ strict-core template inequalities are not built into the stationary kernel; they are part of a rare event E_j^{two}, with their probability cost included in β_{two}.

The spacing ledger must be nonempty. In the basic block setup one needs

$$b < \chi < 1 - \beta_{\text{two}},$$

equivalently $\beta_{\text{two}} + b < 1$. If the truncated local window has exponent λ_{loc} larger than b, the corresponding feasibility condition is

$$\beta_{\text{two}} + \lambda_{\text{loc}} < 1.$$

4 Exterior Coordinates

Use the reciprocal polynomial

$$G_m(y) := y^m f_m(1/y) = \sum_{r=0}^{m} \epsilon_{m-r} y^r.$$

Exterior roots of f_m correspond to roots of G_m in $(-1,0)\cup(0,1)$. Write

$$y=\sigma e^{-t}, \qquad \sigma\in\{\pm 1\}.$$

Then

$$G_m(\sigma e^{-t})=\sum_{r=0}^{m}\epsilon_{m-r}\sigma^r e^{-rt}.$$

At checkpoint n_j, define $\delta_k=\epsilon'_k-\epsilon_k$. The own-block perturbation is

$$\Delta_{j,\sigma}(t)=\sum_{r\in[R_0,R_1]}\sigma^r\delta_{n_j-r}e^{-rt}.$$

Equivalently, in the reciprocal variable,

$$\widetilde{\Delta}_{j,\sigma}(y)=\sum_{r\in[R_0,R_1]}\sigma^r\delta_{n_j-r}y^r, \qquad \widetilde{\Delta}_{j,\sigma}(e^{-t})=\Delta_{j,\sigma}(t).$$

Other switched blocks are treated as tail perturbations. Earlier blocks enter the reciprocal expansion at distances n_j-n_k+R. Since the spacing is $\gtrsim N^{\chi}$ and $\chi>b$, they are exponentially damped in the strict core $t\gtrsim R_1^{-1}$. Later blocks do not enter G_{n_j} after the same spacing choice. In very-near-boundary regimes, where exponential damping is unavailable, the interference lemma must control the cumulative translated moment-matching remainder after summing over earlier checkpoints, using moment cancellation, endpoint guards, and binomial translation stability.

5 Local Two-Endpoint Event

The core local object is a two-endpoint event E_j^{two}. It is an event for the coupled baseline and perturbation data

$$(G_{j,-},\Delta_{j,-};G_{j,+},\Delta_{j,+}),$$

not just for the perturbation. It includes baseline and transversality conditions needed to turn the perturbation into a signed root-count discrepancy, and it simultaneously damps the positive endpoint.

The desired probability estimate is

$$\mathbb{P}(E_j^{\mathrm{two}})=N^{-\beta_{\mathrm{two}}+o(1)}, \qquad \beta_{\mathrm{two}}<1-\chi,$$

uniformly in j, away from harmless boundary trimming. The exponent β_{two} must include all costs:

$$\beta_{\mathrm{two}}=\beta_{\mathrm{fiber}}+\beta_{\mathrm{baseline}}+\beta_{\mathrm{trans}}+\beta_{\mathrm{positive}}+\beta_{\mathrm{regularity}}+\beta_{\mathrm{central}}.$$

In particular it includes the entropy cost that the uniformly sampled same-fiber output realizes a useful perturbation profile, not only the probability of a favorable input fiber. A complete proof must provide a concrete feasible budget with $\beta_{\mathrm{two}}<1-\chi$.

On E_j^{two}, plus the relevant local regularity event, the deterministic realization lemma should give

$$D_{\mathcal{C}_-}\geq\varepsilon_{\mathrm{core}}\log N, \qquad |D_{\mathcal{C}_+}|\leq\frac{1}{4}\varepsilon_{\mathrm{core}}\log N.$$

Here $0<\varepsilon_{\mathrm{core}}\ll b-a$ is fixed after the strict-core cell budget is chosen.

6 Local/Tail Separation and Checkpoint Selection

Split the good event into

$$H_j = H_j^{\mathrm{loc}} \cap H_j^{\mathrm{tail}}.$$

The local event H_j^{loc} contains endpoint, shoulder, and local regularity controls supported on the truncated coefficient window used by the template. The tail event H_j^{tail} contains far-tail and cross-checkpoint interference controls.

For the dependency-graph step, Y_j^{loc} is taken measurable with respect to a coefficient window of size $N^{\lambda_{\mathrm{loc}}+o(1)}$, with $\lambda_{\mathrm{loc}} < \chi$. In the basic own-block model $\lambda_{\mathrm{loc}} = b$, up to endpoint guards.

The local off-core estimate must be relative to the rare template event:

$$\mathbb{P}((H_j^{\mathrm{loc}})^c \mid E_j^{\mathrm{two}}) = o(1).$$

Set

$$Y_j^{\mathrm{loc}} := \mathbf{1}_{E_j^{\mathrm{two}} \cap H_j^{\mathrm{loc}}}.$$

The second-moment step needs

$$\sum_j \mathbb{P}(Y_j^{\mathrm{loc}}) = N^{1-\chi-\beta_{\mathrm{two}}+o(1)} \to \infty$$

and absolute covariance control

$$\sum_{j \neq k} \left| \mathbb{P}(Y_j^{\mathrm{loc}} Y_k^{\mathrm{loc}}) - \mathbb{P}(Y_j^{\mathrm{loc}}) \mathbb{P}(Y_k^{\mathrm{loc}}) \right| = o\left(\left(\sum_j \mathbb{P}(Y_j^{\mathrm{loc}}) \right)^2 \right).$$

The tail part is handled by the selected-success-weighted estimate

$$\sum_j \mathbb{P}\big(Y_j^{\mathrm{loc}} \cap (H_j^{\mathrm{tail}})^c\big) = o\left(\sum_j \mathbb{P}(Y_j^{\mathrm{loc}}) \right).$$

If

$$S_N = \sum_j Y_j^{\mathrm{loc}}, \qquad B_N = \sum_j Y_j^{\mathrm{loc}} \mathbf{1}_{(H_j^{\mathrm{tail}})^c},$$

then the covariance estimate gives $S_N / \mathbb{E} S_N \to 1$ in probability, and the selected-success tail estimate gives $B_N / \mathbb{E} S_N \to 0$. Hence, with probability $1 - o(1)$, some locally successful checkpoint is also tail-good.

7 Finite-Window Linear Algebra

After scaling $x = r/R_1$, the observation functions have the form

$$x^q e^{-\lambda x}$$

with λ in finitely many adjacent geometric progressions. The Riesz bound is needed on the moving window

$$x \in [\rho_N, 1], \qquad \rho_N = N^{a-b} = N^{-\delta},$$

with

$$M \asymp \log(1/\rho_N) \asymp \delta \log N, \qquad \lambda \in [K_0, K_1/\rho_N],$$

where $K_0 > 0$ is fixed after choosing the strict-core guard. The desired uniform estimate is

$$c \sum_\nu |a_\nu|^2 \le \left\| \sum_\nu a_\nu \phi_\nu \right\|^2_{L^2([\rho_N,1])} \le C \sum_\nu |a_\nu|^2,$$

with c, C independent of N, ρ_N, M. The actual theorem is needed in the normalized discrete block norm on $r \in [R_0, R_1]$, with the continuous form serving as the model after a uniform discretization comparison. The fact that $R_0 = N^a \to \infty$ keeps the lowest geometric cells populated.

For the two-endpoint template, the combined ordinary/alternating observation family has representatives

$$\phi_{\sigma,\nu}(r) = \sigma^r x^q e^{-\lambda_\nu x}, \qquad \sigma \in \{\pm 1\}.$$

The Riesz and projected-Schur estimates are needed for this combined endpoint family, not only for one parity sector.

The Schur complement direction needed by same-fiber perturbations is the projected-observation quotient form. Let

$$S_A = \operatorname{span}\{r^q, (-1)^r r^q : 0 \le q \le A\}.$$

Since the perturbation lies in the kernel of the moment map, the projected strict-core observation family $P_{S_A^\perp}\phi_\nu$ must remain uniformly Riesz in the normalized discrete block norm. Equivalently,

$$G_V - B G_S^{-1} B^T \ge c\, G_V$$

uniformly in N, ρ_N, M. A reverse Schur bound excluding approximation of moment modes by observations may also be useful, but this projected quotient form is the load-bearing one for fiber-conditioned template realization.

8 Endpoint and Shoulder Stability

Normalize at checkpoint n_j by

$$X_{\sigma,j}(s) = \frac{G_{n_j}(\sigma e^{-s})}{\sqrt{\mathcal{V}_j(s)}}, \qquad h^{\text{own}}_{\sigma,j}(s) = \frac{\Delta_{j,\sigma}(s)}{\sqrt{\mathcal{V}_j(s)}} = \frac{\widetilde{\Delta}_{j,\sigma}(e^{-s})}{\sqrt{\mathcal{V}_j(s)}},$$

where

$$\mathcal{V}_j(s) = \sum_{r=0}^{n_j} e^{-2rs}.$$

The own-block homotopy is

$$F^{\text{own}}_{\tau,j,\sigma}(s) = X_{\sigma,j}(s) + \tau h^{\text{own}}_{\sigma,j}(s).$$

The actual perturbation is

$$h^{\text{tot}}_{\sigma,j} = h^{\text{own}}_{\sigma,j} + h^{\text{tail}}_{\sigma,j}.$$

The endpoint/shoulder proof must either apply directly to $X_{\sigma,j} + \tau h^{\text{tot}}_{\sigma,j}$, or proceed in two steps: own-block first, then tail perturbation at $o(\log N)$ zero-count cost.

Along the own-block homotopy, zero-count changes occur only through boundary crossings or double roots. On the regular event,

$$\left|\det \nabla(F^{\mathrm{own}}_{\tau,j,\sigma}, \partial_s F^{\mathrm{own}}_{\tau,j,\sigma})\right| = |h^{\mathrm{own}}_{\sigma,j}\, \partial_s^2 F^{\mathrm{own}}_{\tau,j,\sigma}|.$$

This is the fold-count mechanism behind the off-core stability estimate.

For Rademacher signs this is not literal Kac-Rice. The needed replacement is a polynomial-scale conditional anti-concentration theorem under the actual fiber/template law. For admissible conditioning Θ_j, for s in off-core endpoint or shoulder ranges not already constrained by the core template, for $\tau \in [0,1]$, and for $r \in [N^{-B}, 1]$,

$$\mathbb{P}\left(|F^{\mathrm{own}}_{\tau,j,\sigma}(s)| \le r,\ |\partial_s F^{\mathrm{own}}_{\tau,j,\sigma}(s)| \le r \mid \Theta_j\right) \le Cr^2 + N^{-\gamma_{\mathrm{tr}}} r^{-C_*},$$

with the corresponding one-dimensional endpoint estimate

$$\mathbb{P}\left(|F^{\mathrm{own}}_{\tau,j,\sigma}(s_*)| \le r \mid \Theta_j\right) \le Cr + N^{-\gamma_{\mathrm{tr}}} r^{-C_*}.$$

The smoothing exponent must satisfy

$$0 < B < \gamma_{\mathrm{tr}}/(C_* + 3).$$

The word admissible means that, after conditioning on finite fiber data and the $O(\log N)$-cell core template, every off-core endpoint or shoulder jet still has reserve independent, or quantitatively unstructured, signs. Non-admissible fibers are charged to the central-fiber/regularity cost or to H_j^{loc}. In core cells deliberately constrained by E_j^{two}, anti-concentration is replaced by the deterministic local realization lemma.

9 Inside Interval and Boundary Roots

Do's theorem gives, for both iid marginals,

$$\sup_{N \le m \le 2N} |N_m([-1,1]; f) - N_m([-1,1]; f')| = o_{\mathbb{P}}(\log N).$$

This is a global dyadic high-probability event. It is not conditioned on the rare checkpoint. After the exterior-good checkpoint is selected, one intersects with this global event. Roots exactly at $x = \pm 1$ are assigned to the inside interval and counted only once.

10 Assembly

On the selected event,

$$D_{\mathcal{C}_-} \ge \varepsilon_{\mathrm{core}} \log N, \qquad |D_{\mathcal{C}_+}| \le \frac{1}{4}\varepsilon_{\mathrm{core}} \log N,$$

and

$$|D_{\mathrm{endpoint/shoulder/far}}| + |D_{\mathrm{inside}}| \le \frac{1}{4}\varepsilon_{\mathrm{core}} \log N.$$

Therefore

$$R_{n_j}(f) - R_{n_j}(f') = D_{\mathcal{C}_-} + D_{\mathcal{C}_+} + D_{\mathrm{endpoint/shoulder/far}} + D_{\mathrm{inside}}$$

has magnitude at least $c_0 \log N$ for some fixed $c_0 > 0$. This produces the forbidden dyadic coupling discrepancy and contradicts the any-coupling consequence of the assumed almost-sure law.

11 Conditional Inputs Checklist

The roadmap is conditional on the following inputs.

1. Finite-window Mellin/Riesz on $x \in [\rho_N, 1]$, in the actual normalized discrete block norm, uniform as $\rho_N \to 0$ and $M \asymp \log(1/\rho_N)$.

2. Projected-observation quotient Schur after imposing ordinary/alternating moment constraints:

$$G_V - BG_S^{-1}B^T \geq cG_V.$$

3. Reverse balanced Schur, if used, against $\operatorname{span}\{r^q, (-1)^r r^q : 0 \leq q \leq A\}$.

4. Two-endpoint template probability

$$\mathbb{P}(E_j^{\text{two}}) = N^{-\beta_{\text{two}}+o(1)}, \qquad \beta_{\text{two}} < 1 - \chi,$$

including baseline, transversality, positive-core damping, same-fiber entropy, central-fiber regularity, and all local regularity costs.

5. Blockwise doubly stochastic switching preserving the full iid Rademacher product law.

6. Central-fiber regularity/local-CLT control, including typical fiber sizes, lattice-span and periodicity issues, and conditional small-ball estimates under admissible rare template data.

7. Off-core/template reserve-randomness Schur or transversality, ensuring that off-core jets retain enough residual randomness after the core template is conditioned.

8. Conditional off-core stability under the full rare event:

$$\mathbb{P}((H_j^{\text{loc}})^c \mid E_j^{\text{two}}) = o(1).$$

9. Cross-checkpoint interference control, with exponential damping in the strict core and cumulative translated-moment control near the boundary.

10. Pairwise decorrelation/dependency-graph estimates for truncated local checkpoint variables Y_j^{loc}.

11. Selected-success tail control:

$$\sum_j \mathbb{P}(Y_j^{\text{loc}} \cap (H_j^{\text{tail}})^c) = o\left(\sum_j \mathbb{P}(Y_j^{\text{loc}})\right).$$

12. Polynomial-scale conditional anti-concentration/sign transfer for endpoint triple jets and shoulder double zeros under the actual fiber/template conditioning, with errors of the form

$$Cr^d + N^{-\gamma_{\text{tr}}} r^{-C_*}$$

down to $r = N^{-B}$.

13. Endpoint homotopy/fold-count replacement for Kac-Rice in the sign model, applied either to the total perturbation or to the own-block perturbation plus a tail-stability step.

14. Final assembly with Do's global inside-good event and consistent bookkeeping of roots at $x = \pm 1$.

12 Current Status

The architecture is intended to be a reviewable conditional roadmap, not a proof announcement. The global coupling contradiction, iid-marginal preservation, reciprocal exterior formulation, local/tail selection logic, and inside-interval bookkeeping are structurally fixed. The real mathematical burden is now concentrated in the support lemmas above, especially the two-endpoint exponent budget, central-fiber regularity, moving-window Riesz/Schur estimates, conditional sign transfer, and cumulative boundary interference.

Process Note

This roadmap was developed with a self-drive research-loop workflow in Codex, using ledgers of targets, failed routes, and handoffs, and then GPT Pro was used for external pressure-testing. That workflow is secondary to the mathematics; I am happy to share it separately if useful.